Machine Learning Lab – Question 17

Question

Implement Mobile Price Prediction using an appropriate machine learning algorithm.

Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

data=pd.DataFrame({
"ram":[2,3,4,4,6,8,8,12,2,3,6,8],
"battery":[3000,3500,4000,4500,4500,5000,5200,6000,2800,3600,4800,5100],
"storage":[32,64,64,128,128,256,256,256,32,64,128,256],
"camera":[8,12,16,20,32,48,50,108,8,13,32,64],
"price_range":[0,0,1,1,2,3,3,3,0,1,2,3]
})
X=data.drop(columns="price_range"); y=data.price_range
X_train,X_test,y_train,y_test=train_test_split(
    X,y,test_size=.25,random_state=42,stratify=y)

model=RandomForestClassifier(n_estimators=100,random_state=42)
model.fit(X_train,y_train)
print("Accuracy:",round(accuracy_score(y_test,model.predict(X_test)),4))
print("Predicted price range:",model.predict([[8,5000,256,50]])[0])
```

Output

```
The program prints classification accuracy and predicted price range (0–3)
for the supplied phone specification.
```